■ ReconAI: 4-Minute Demo Video Script

End-Term Software Project • Track 04: AI Finance Controller • **Author:** Mahendra Varanasi

Target Duration: Exact 4 Minutes (~510 words)	**Live Demo:** reconai-ogbb8gwzyyhmebevo3nr79.streamlit.app	**Rubric (Sec 14):** 100% Product Walkthrough • 0% Code

Time	Section 14 Rubric Requirement	Screen Action / Interaction
0:00 – 0:45	The Problem & Target Audience	Show app top header on live Streamlit Cloud
0:45 – 1:45	Major Workflow 1: Deterministic Engine	Show 9 KPI cards & Section 3 Breakdown table in Demo Mode
1:45 – 2:45	Most Interesting Feature: Gemini AI Escalation	Section 5 Drill-Down: select TXN0005 (match) & TXN0042 (safety)
2:45 – 3:30	Major Workflow 2: Live In-Memory Upload	Switch to Upload Mode, click 3 download buttons, drop files & Run
3:30 – 4:00	Final Working App & Ground Truth Oracle	Section 4 Confusion Matrix (97.8% Accuracy, 0 False Positives)

Part 1: The Problem & Target Audience (0:00 – 0:45)

■■ SCREEN ACTION: Open browser to live app (https://reconai-ogbb8gwzyyhmebevo3nr79.streamlit.app/). Show top header and overview.

“Hello everyone. My name is **Mahendra Varanasi**, and this is **ReconAI**, an autonomous multi-source financial reconciliation engine.

Who is this for? This product is built for finance controllers, accounting teams, and fintech companies managing high-volume transactions.

What problem are we solving? In businesses, money moves across three separate systems: **Payment Gateways**, **Bank Settlement Statements**, and the **Internal Accounting Ledger**. Reconciling these manually is painful because bank settlements take 1 to 3 days to arrive, gateways deduct 2% fees, reference numbers get truncated, and duplicate entries happen.

Manual Excel auditing takes days, while basic fuzzy matching creates dangerous false matches. ReconAI solves this with a clear architecture: **deterministic rules first, Gemini AI escalation second, and zero false positives.**”

Part 2: Major Workflow 1 — Deterministic Matching (Demo Mode) (0:45 – 1:45)

■■ SCREEN ACTION: Scroll down slightly to display the 9 Executive KPI cards, then point cursor to Section 3: Reconciliation Breakdown table.

“Let us look at our first major workflow: **Demo Mode**, running on a 90-transaction benchmark dataset.

Out of 90 transactions, our deterministic rule engine immediately resolves **78 transactions** with zero external API calls:
• **68 transactions are confirmed matches:** 60 were exact matches across dates and names, and 8 were matched automatically by recognizing the standard 2% gateway fee deduction within the 3-day settlement window.
• **10 transactions are caught as rule exceptions:** 5 duplicate settlements where multiple bank entries existed, and 5 missing records where no settlement arrived.

Deterministic rules handle about 85% of transactions at zero cost, ensuring mathematical accuracy without guessing.”

Part 3: Most Interesting Feature — Selective AI Escalation & Safety (1:45 – 2:45)

■■ SCREEN ACTION: Scroll down to Section 5: Transaction Drill-Down Explorer. Select TXN0005, show Gemini reasoning, then select TXN0042.

"Now, for the most interesting feature: **Selective AI Escalation**.

What happens to the remaining **12 ambiguous cases**? Instead of dumping our whole database to an LLM, ReconAI extracts only the conflicting candidate records and sends them to **Google Gemini** with strict Pydantic JSON schema validation.

Let us look at the **Transaction Drill-Down**:
• *(Select TXN0005)* In **TXN0005**, two identical bank entries existed for the same customer and amount. Our rules refused to guess. But Gemini inspected the cryptic bank description, found the matching reference code, and correctly resolved it with **98% confidence**.
• *(Select TXN0042)* Now look at **TXN0042**. What happens if the AI encounters a rate limit or API failure? ReconAI does not crash, and it never forces a fake match. It safely flags it as **AI_FAILED**. In financial systems, flagging an exception is safe; making a wrong match is fatal."

Part 4: Major Workflow 2 — Live In-Memory Upload Mode (2:45 – 3:30)

■■ SCREEN ACTION: In the sidebar, select 'Upload CSV Data'. Expand 'Need sample files to test?', click the 3 download buttons, drop files into uploaders, and click Run Reconciliation.

"Next is our second major workflow: **Live CSV Upload Mode**.

ReconAI is not just a pre-computed demo—it is a live application where users can upload real operational feeds.

For instant testing, we built pre-formatted test CSVs right into the app:
• *(Click 3 download buttons)* I can download the sample Payment, Bank, and Ledger files in one click.
• *(Drag 3 files into uploaders)* The app validates the required column headers and file formats in real time.
• *(Click 'Run Reconciliation')* When I click **Run Reconciliation**, the engine executes completely in-memory, producing instant KPI metrics and breakdown tables for the uploaded files."

Part 5: Final Working Application & Ground-Truth Verification (3:30 – 4:00)

■■ SCREEN ACTION: Switch back to Demo Mode in the sidebar and scroll to Section 4: Evaluation & Ground-Truth Oracle Performance. Point cursor to Confusion Matrix.

"Finally, let us verify system accuracy using our independent **Ground Truth Oracle**:
• **Overall Binary Accuracy:** 97.8%
• **False Positives:** Exactly **0** — our system has a 0.00% False Positive Rate.
• **Precision:** 100% across all exception categories.

The entire application is written in **Python**, using Streamlit for the responsive UI, Google Gemini for reasoning, and deployed live on Streamlit Cloud.

This concludes the demonstration of ReconAI. Thank you!"

■ **Quick Recording Advice:** Speak in a steady, natural tone (~130 words/minute). Move your cursor smoothly to the KPI card or dropdown you are speaking about. Pause 1 second after each click so the viewer can clearly see the action.